

Appendix B. Patient-Reported Outcome Measures Questionnaires

36-Item Short Form (SF-36) Health Survey (Version 2)

The physical component summary (PCS) score, derived from the 36-Item Short Form (SF-36) Health Survey (Version 2) was the primary outcome. The range of the SF-36 PCS is between 0 and 100, where higher scores represent better physical functioning.

Your Health and Well-Being

This survey asks for your views about your health. This information will help keep track of how you feel and how well you are able to do your usual activities. Thank you for completing this survey!

For each of the following questions, please mark an ☐ in the one box that best describes your answer.

1. In general, would you say your health is:

Excellent	Very good	Good	Fair	Poor
▼	▼	▼	▼	▼
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

2. Compared to one year ago, how would you rate your health in general now?

Much better now than one year ago	Somewhat better now than one year ago	About the same as one year ago	Somewhat worse now than one year ago	Much worse now than one year ago
▼	▼	▼	▼	▼
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

3. The following questions are about activities you might do during a typical day. Does your health now limit you in these activities? If so, how much?

	Yes, limited a lot ▼	Yes, limited a little ▼	No, not limited at all ▼
a <u>Vigorous activities</u> , such as running, lifting heavy objects, participating in strenuous sports	☐ 1	☐ 2	☐ 3
b <u>Moderate activities</u> , such as moving a table, pushing a vacuum cleaner, bowling, or playing golf	☐ 1	☐ 2	☐ 3
c Lifting or carrying groceries.....	☐ 1	☐ 2	☐ 3
d Climbing <u>several</u> flights of stairs.....	☐ 1	☐ 2	☐ 3
e Climbing <u>one</u> flight of stairs	☐ 1	☐ 2	☐ 3
f Bending, kneeling, or stooping	☐ 1	☐ 2	☐ 3
g Walking <u>more than a mile</u>	☐ 1	☐ 2	☐ 3
h Walking <u>several hundred yards</u>	☐ 1	☐ 2	☐ 3
i Walking <u>one hundred yards</u>	☐ 1	☐ 2	☐ 3
j Bathing or dressing yourself	☐ 1	☐ 2	☐ 3

4. During the past 4 weeks, how much of the time have you had any of the following problems with your work or other regular daily activities as a result of your physical health?

	All of the time	Most of the time	Some of the time	A little of the time	None of the time
a Cut down on the <u>amount of time</u> you spent on work or other activities	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
b <u>Accomplished less</u> than you would like	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
c Were limited in the <u>kind of</u> work or other activities	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
d Had <u>difficulty</u> performing the work or other activities (for example, it took extra effort)	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

5. During the past 4 weeks, how much of the time have you had any of the following problems with your work or other regular daily activities as a result of any emotional problems (such as feeling depressed or anxious)?

	All of the time	Most of the time	Some of the time	A little of the time	None of the time
a Cut down on the <u>amount of time</u> you spent on work or other activities	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
b <u>Accomplished less</u> than you would like	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
c Did work or other activities <u>less carefully than usual</u>	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

6. During the past 4 weeks, to what extent has your physical health or emotional problems interfered with your normal social activities with family, friends, neighbors, or groups?

Not at all	Slightly	Moderately	Quite a bit	Extremely
▼	▼	▼	▼	▼
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

7. How much bodily pain have you had during the past 4 weeks?

None	Very mild	Mild	Moderate	Severe	Very severe
▼	▼	▼	▼	▼	▼
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6

8. During the past 4 weeks, how much did pain interfere with your normal work (including both work outside the home and housework)?

Not at all	A little bit	Moderately	Quite a bit	Extremely
▼	▼	▼	▼	▼
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

9. These questions are about how you feel and how things have been with you during the past 4 weeks. For each question, please give the one answer that comes closest to the way you have been feeling. How much of the time during the past 4 weeks...

	All of the time	Most of the time	Some of the time	A little of the time	None of the time
	▼	▼	▼	▼	▼
a Did you feel full of life?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
b Have you been very nervous?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
c Have you felt so down in the dumps that nothing could cheer you up?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
d Have you felt calm and peaceful?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
e Did you have a lot of energy?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
f Have you felt downhearted and depressed?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
g Did you feel worn out?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
h Have you been happy?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
i Did you feel tired?	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

10. During the past 4 weeks, how much of the time has your physical health or emotional problems interfered with your social activities (like visiting with friends, relatives, etc.)?

All of the time	Most of the time	Some of the time	A little of the time	None of the time
▼	▼	▼	▼	▼
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

11. How TRUE or FALSE is each of the following statements for you?

	Definitely true	Mostly true	Don't know	Mostly false	Definitely false
	▼	▼	▼	▼	▼
a I seem to get sick a little easier than other people	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
b I am as healthy as anybody I know	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
c I expect my health to get worse.....	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
d My health is excellent.....	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

Thank you for completing these questions!

Neck Disability Index (NDI)

The Neck Disability Index (NDI) measures how neck pain affects the patients' ability to manage in everyday life. Each section is scored on a scale of 0 to 5, where 0 = "no pain" and 5 = "worst imaginable pain." The summed total range is between 0 and 100, with a lower score representing less disability.

Please Read: This questionnaire is designed to enable us to understand how much your neck pain has affected your ability to manage everyday activities. Please answer each Section by circling the **ONE CHOICE** that most applies to you. We realize that you may feel that more than one statement may relate to you, but Please **just circle the one choice which closely describes your problem right now.**

SECTION 1--Pain Intensity

- A. I have no pain at the moment
- B. The pain is mild at the moment.
- C. The pain comes and goes and is moderate.
- D. The pain is moderate and does not vary much.
- E. The pain is severe but comes and goes.
- F. The pain is severe and does not vary much.

SECTION 2--Personal Care (Washing, Dressing etc.)

- A. I can look after myself without causing extra pain.
- B. I can look after myself normally but it causes extra pain.
- C. It is painful to look after myself and I am slow and careful.
- D. I need some help, but manage most of my personal care.
- E. I need help every day in most aspects of self-care.
- F. I do not get dressed, I wash with difficulty and stay in bed.

SECTION 3--Lifting

- A. I can lift heavy weights without extra pain.
- B. I can lift heavy weights, but it causes extra pain.
- C. Pain prevents me from lifting heavy weights off the floor but I can if they are conveniently positioned, for example on a table.
- D. Pain prevents me from lifting heavy weights, but I can manage light to medium weights if they are conveniently positioned.
- E. I can lift very light weights.
- F. I cannot lift or carry anything at all.

SECTION 4 --Reading

- A. I can read as much as I want to with no pain in my neck.
- B. I can read as much as I want with slight pain in my neck.
- C. I can read as much as I want with moderate pain in my neck.
- D. I cannot read as much as I want because of moderate pain in my neck.
- E. I cannot read as much as I want because of severe pain in my neck.
- F. I cannot read at all.

SECTION 5--Headache

- A. I have no headaches at all.
- B. I have slight headaches which come infrequently.
- C. I have moderate headaches which come in-frequently.
- D. I have moderate headaches which come frequently.
- E. I have severe headaches which come frequently.
- F. I have headaches almost all the time.

SECTION 6 -- Concentration

- A. I can concentrate fully when I want to with no difficulty.
- B. I can concentrate fully when I want to with slight difficulty.
- C. I have a fair degree of difficulty in concentrating when I want to.
- D. I have a lot of difficulty in concentrating when I want to.
- E. I have a great deal of difficulty in concentrating when I want to.
- F. I cannot concentrate at all.

SECTION 7--Work

- A. I can do as much work as I want to.
- B. I can only do my usual work, but no more.
- C. I can do most of my usual work, but no more.
- D. I cannot do my usual work.
- E. I can hardly do any work at all.
- F. I cannot do any work at all.

SECTION 8--Driving

- A. I can drive my car without neck pain.
- B. I can drive my car as long as I want with slight pain in my neck.
- C. I can drive my car as long as I want with moderate pain in my neck.
- D. I cannot drive my car as long as I want because of moderate pain in my neck.
- E. I can hardly drive my car at all because of severe pain in my neck.
- F. I cannot drive my car at all.

SECTION 9--Sleeping

- A. I have no trouble sleeping
- B. My sleep is slightly disturbed (less than 1 hour sleepless).
- C. My sleep is mildly disturbed (1-2 hours sleepless).
- D. My sleep is moderately disturbed (2-3 hours sleepless).
- E. My sleep is greatly disturbed (3-5 hours sleepless).
- F. My sleep is completely disturbed (5-7 hours sleepless).

SECTION 10--Recreation

- A. I am able engage in all recreational activities with no pain in my neck at all.
- B. I am able engage in all recreational activities with some pain in my neck.
- C. I am able engage in most, but not all recreational activities because of pain in my neck.
- D. I am able engage in a few of my usual recreational activities because of pain in my neck.
- E. I can hardly do any recreational activities because of pain in my neck.
- F. I cannot do any recreational activities all all.

Subject ID: _____ Visit: _____

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DISABILITY INDEX SCORE: % _____

Modified Japanese Orthopedic Association (mJOA)

The modified Japanese Orthopedic Association (mJOA) score is used to evaluate the functional status of patients with degenerative cervical myelopathy. Each section is summed to obtain a score (range = 0-17), with a higher score representing less dysfunction from myelopathy.⁵ A typical patient with moderate cervical myelopathy has a mJOA score between 12 and 14. Many other surgical studies show that patients with cervical myelopathy have mJOA scores in this range.

mJOA Scale

mJOA: _____ (0-17)

Motor, arms

- 0 Unable to feed oneself
- 1 Unable to use a knife and fork, able to eat with spoon
- 2 Able to use knife and fork with much difficulty
- 3 Able to use knife and fork with slight difficulty
- 4 No deficit

Motor, legs

- 0 Unable to walk
- 1 Can walk on flat floor with a walking aid
- 2 Can walk up or down stairs with a handrail
- 3 Lack of stability and smooth gait
- 4 No deficit

Sensation, arms

- 0 Severe sensory loss or pain
- 1 Mild sensory loss
- 2 No deficit

Sensation, legs

- 0 Severe sensory loss or pain
- 1 Mild sensory loss
- 2 No deficit

Sensation, trunk

- 0 Severe sensory loss or pain
- 1 Mild sensory loss
- 2 No deficit

Bladder function

- 0 Unable to void
- 1 Marked difficulty with micturation (retention)
- 2 Difficulty in micturation (frequency, hesitation)
- 3 No deficit

EuroQol 5-Dimensions (EQ-5D)

The 3-level version of the EuroQol-5 Dimensions (EQ-5D) comprises of five discrete aspects of quality-of-life that taken together, sum into one number (between 0 and 1), where the higher score denotes greater quality-of-life. The second aspect of the EQ-5D is a visual analogue scale (VAS) to assess the patients' current health between the best state (100) and worst state imaginable (0).



Health Questionnaire
(English version for the US)

Subject ID _____

Visit _____

By placing a checkmark in one box in each group below, please indicate which statements best describe your own health state today.

Mobility

- I have no problems in walking about ☐
- I have some problems in walking about ☐
- I am confined to bed ☐

Self-Care

- I have no problems with self-care ☐
- I have some problems washing or dressing myself ☐
- I am unable to wash or dress myself ☐

Usual Activities (*e.g. work, study, housework, family or leisure activities*)

- I have no problems with performing my usual activities ☐
- I have some problems with performing my usual activities ☐
- I am unable to perform my usual activities ☐

Pain/Discomfort

- I have no pain or discomfort ☐
- I have moderate pain or discomfort ☐
- I have extreme pain or discomfort ☐

Anxiety/Depression

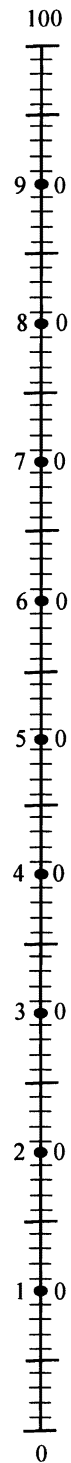
- I am not anxious or depressed ☐
- I am moderately anxious or depressed ☐
- I am extremely anxious or depressed ☐

To help people say how good or bad a health state is, we have drawn a scale (rather like a thermometer) on which the best state you can imagine is marked 100 and the worst state you can imagine is marked 0.

We would like you to indicate on this scale how good or bad your own health is today, in your opinion. Please do this by drawing a line from the box below to whichever point on the scale indicates how good or bad your health state is today.

**Your own
health state
today**

Best
imaginable
health state



Worst
imaginable
health state